

TryNex Lifestyle (trynexshop.com) — Full Audit Report

Executive Summary

A comprehensive deep-dive audit was conducted on trynexshop.com, a print-on-demand e-commerce platform built with Next.js/React. The site offers custom product design (T-shirts, Hoodies, Mugs, Caps) with an interactive Design Studio. Critical issues were found across the Design Studio, Frontend Responsiveness, Admin Panel, and Backend connectivity. This report documents all findings and their root causes.

Site Overview

Attribute	Details
URL	trynexshop.com
Tech Stack	Next.js / React
CDN/Security	Cloudflare
Products	T-Shirts, Hoodies, Mugs, Caps
Currency	BDT (৳)
Key Feature	Design Studio (/customize)
Fabric Quality	Premium 320GSM
Free Delivery	Orders above ৳1500

Issue Category 1: Design Studio (/customize)

1.1 Broken Interaction System

Severity: CRITICAL

The Design Studio's touch/mouse interaction system is fundamentally broken. Users cannot:

- Click and select uploaded designs

- Hold and drag designs to reposition them
- Resize designs by dragging corners
- Deselect designs by clicking outside the print area

Root Cause: The implementation likely uses `onClick` / `onMouseDown` events only, which do not translate to touch devices. There is no unified pointer event handling, and `touch-action` CSS is not properly configured on the canvas container.

1.2 Incorrect Print Areas

Severity: CRITICAL

Every product mockup displays an incorrect print area. The borders showing where the design will be printed do not match the actual physical dimensions of the product's printable surface.

Root Cause: The print area coordinates (x, y, width, height) stored in the configuration do not correspond to the actual product specifications. The aspect ratios are wrong — for example, a T-shirt front print area should be approximately 12" × 16" (3:4 ratio), but the displayed area appears to have incorrect proportions.

1.3 Auto-Fit Failure

Severity: HIGH

When users upload an image, it does not automatically scale and center within the print area. Images either overflow the boundaries, appear too small, or lose their aspect ratio.

Root Cause: Missing or broken auto-fit calculation that should compute the optimal scale based on the image dimensions relative to the print area dimensions.

1.4 Color Change Sizing Bug

Severity: HIGH

When users change the product color (e.g., from white to black T-shirt), the entire mockup container changes size — sometimes becoming larger, sometimes smaller. This causes layout shifts and disrupts any placed designs.

Root Cause: The mockup container does not have a fixed aspect ratio. Different color variant images may have slightly different dimensions, and without `aspect-ratio` CSS or fixed container sizing, the layout reflows on each color change.

1.5 Design Overlay Issue

Severity: MEDIUM

Uploaded designs appear to float on top of the product rather than looking like they are actually printed on the fabric. There is no visual integration between the design and the product texture.

Root Cause: Missing CSS `mix-blend-mode` (multiply for light products, screen for dark) and no product texture overlay layer.

Issue Category 2: Frontend Responsiveness

2.1 Global Scroll Issues

Severity: HIGH

- Desktop: Mouse wheel scrolling is broken or janky in multiple areas
- Mobile: Touch scrolling doesn't work properly in certain sections
- The Design Studio page may block page-level scrolling entirely

Root Cause: Likely `overflow: hidden` applied to parent containers, or a custom scroll library interfering with native scroll behavior. The Design Studio's `touch-action: none` (if present) may be applied too broadly, blocking page scroll.

2.2 Horizontal Overflow

Severity: HIGH

Pages exhibit horizontal scrolling on mobile devices, indicating elements that exceed the viewport width.

Root Cause: Fixed-width elements, images without `max-width: 100%`, or absolute-positioned elements extending beyond the viewport.

2.3 Layout Breakpoints

Severity: MEDIUM

The responsive design does not properly adapt between mobile, tablet, and desktop breakpoints. Product grids, navigation, and content sections do not fluidly transition.

Root Cause: Insufficient media queries or reliance on fixed pixel widths instead of fluid/percentage-based layouts.

Issue Category 3: Admin Panel

3.1 Database Connection Inconsistency

Severity: HIGH

The admin dashboard displays "Database: Not Connected" in some areas while showing "Connected" in others. This creates confusion about the actual system status.

Root Cause: Multiple possible causes:

- 1. Serverless cold-start connection timeouts (common with Prisma on serverless platforms)
- 2. Multiple database status checks using different connection methods
- 3. Environment variable DATABASE_URL not consistently available across all API routes
- 4. Missing connection pooling (each serverless function creates a new connection)

3.2 Admin Responsiveness

Severity: MEDIUM

The admin panel is not responsive on mobile/tablet devices. Tables overflow, sidebars don't collapse, and forms are difficult to use on smaller screens.

Issue Category 4: Performance & Quality

4.1 Console Errors

JavaScript errors are present in the browser console, indicating unhandled exceptions that may contribute to the interaction failures.

4.2 Image Loading

Product images and mockups may not be using Next.js <Image /> component, leading to unoptimized loading, layout shifts (CLS), and inconsistent sizing.

4.3 UI Inconsistency

The overall design lacks a cohesive design system — inconsistent spacing, typography, shadows, and interaction patterns across different pages.

Benchmarking Against Industry Standards

Feature	Printful/Printify	TryNexShop (Current)
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Design drag/move	Smooth, real-time	Broken
Print area accuracy	Precise, with safe-area guides	Incorrect dimensions
Auto-fit on upload	Intelligent scaling	Not working
Color change stability	No layout shift	Causes sizing bugs
Design realism	Blend modes + texture	Floating overlay
Mobile responsiveness	Fully responsive	Broken in multiple areas
Scroll behavior	Native, smooth	Blocked/janky
Admin panel	Responsive dashboard	Desktop-only, DB issues

Recommended Fix Priority

1. **CRITICAL** — Design Studio interaction system (pointer events)
2. **CRITICAL** — Print area coordinate correction
3. **HIGH** — Global scroll and overflow fixes
4. **HIGH** — Auto-fit algorithm implementation
5. **HIGH** — Color change container stability
6. **HIGH** — Database connection pooling
7. **MEDIUM** — Design blend mode realism
8. **MEDIUM** — Page-by-page responsive fixes
9. **MEDIUM** — Admin panel responsiveness
10. **LOW** — Premium UI polish and performance optimization

Conclusion

The TryNex Lifestyle platform has significant technical debt in its Design Studio implementation and overall frontend architecture. The core business feature (custom design placement) is non-functional for most users due to broken interactions and incorrect print areas. A systematic overhaul following the priority order above will bring the platform to production-grade quality matching industry leaders like Printful and Printify.

The accompanying **Master Command Prompt** document provides the exact technical implementation details for an AI coding agent to execute all fixes in a single comprehensive

pass.